REVISION WORKSHEET

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MINISTRY OF EDUCATION

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وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 11 Advanced | Module 8 Lesson 2 | Rotational Dynamics

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: $\tau = rF \sin\theta$; $I = \Sigma mr^2$; $\tau_{net} = I\alpha$; $\alpha = (\omega_f - \omega_i)/\Delta t$. Counterclockwise is positive; neglect rod masses where stated.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 11 book, printed pages 7–17.

P01. Which quantity describes the turning effect of a force about an axis?

A. Linear momentum B. Kinetic energy
C. Torque D. Mass

Answer

C. Torque.

P02. For the same net torque, a smaller moment of inertia gives:

A. A larger angular acceleration B. A smaller angular acceleration
C. The same angular acceleration D. Zero angular velocity

Answer

A. A larger angular acceleration; $\alpha = \tau_{net}/I$.

P03. What is the SI unit of moment of inertia?

A. N m B. kg m/s
C. rad/s² D. kg m²

Answer

D. kg m².

P04. A 20 N tangential force acts 0.30 m from an axle. Its torque magnitude is:

A. 0.015 N m B. 6.0 N m
C. 60 N m D. 66.7 N m

Answer

B. 6.0 N m; $\tau = rF = 0.30 \times 20$.

P05. A force acts along a line that passes through a wheel's axle. Its torque about that axle is:

A. rF B. rF/2
C. Zero D. F/r

Answer

C. Zero; the perpendicular lever arm is zero.

P06. At which angle between radius and force is the torque magnitude greatest?

A. 0° B. 90°
C. 180° D. 360°

Answer

B. 90°, since sin 90° = 1.

P07. A net torque of 12 N m acts on an object with I = 3.0 kg m². Its angular acceleration is:

A. 4.0 rad/s² B. 9.0 rad/s²
C. 15 rad/s² D. 36 rad/s²

Answer

A. 4.0 rad/s²; α = 12/3.0.

P08. Two point masses are each moved from r to 3r from a fixed axis. Their total moment of inertia becomes:

A. Three times the original B. Six times the original
C. One third of the original D. Nine times the original

Answer

D. Nine times the original; I is proportional to r².

P09. If an object has zero net torque about a fixed axis, its angular acceleration is:

A. Always positive B. Zero
C. Always negative D. Equal to its angular velocity

Answer

B. Zero; τnet = Iα. Its angular velocity may be nonzero.

P10. For a fixed force magnitude and radius, which change increases the torque magnitude?

A. Reducing the force B. Halving the radius
C. Changing the angle from 30° to 90° D. Directing the force through the pivot

Answer

C. Increasing the angle from 30° to 90° increases sin θ.

P11. Define torque about an axis.

Answer

The turning effect of a force about an axis.

P12. Define moment of inertia about an axis.

Answer

A measure of resistance to angular acceleration about a specified axis; for point masses, I = Σmr².

P13. Define the perpendicular lever arm of a force.

Answer

The shortest perpendicular distance from the axis or pivot to the force's line of action.

P14. State Newton's second law for rotational motion about a fixed axis.

Answer

The net torque equals moment of inertia times angular acceleration: τnet = Iα.

P15. State the sign of a counterclockwise torque when counterclockwise is chosen as positive.

Answer

Positive.

P16. A 50 N force acts perpendicular to a spanner 0.24 m from its pivot. Calculate the torque magnitude.

Answer

$\tau = rF = 0.24 \times 50 = 12$ N m.

P17. A 30 N force acts at a perpendicular distance of 0.15 m from an axis. Calculate the torque magnitude.

Answer

$\tau = Fl = 30 \times 0.15 = 4.5$ N m.

P18. A point mass of 0.50 kg is 0.40 m from a fixed axis. Calculate its moment of inertia.

Answer

$I = mr^2 = 0.50 \times 0.40^2 = 0.080$ kg m².

P19. Two point masses, 0.20 kg at 0.30 m and 0.40 kg at 0.20 m, rotate about the same fixed axis. Calculate their total moment of inertia.

Answer

$I = 0.20(0.30^2) + 0.40(0.20^2) = 0.018 + 0.016 = 0.034$ kg m².

P20. A disk with $I = 0.50$ kg m² experiences a net torque of 3.0 N m. Calculate its angular acceleration.

Answer

$\alpha = \tau_{net}/I = 3.0/0.50 = 6.0$ rad/s².

P21. A 60 N force acts 0.40 m from a pivot at 30° to the radius. Its turning effect is clockwise.

a) State what is meant by the perpendicular lever arm. [1]

b) Calculate the torque magnitude. [2]

c) State the signed torque using counterclockwise as positive. [1]

Answer

a) Shortest distance from pivot to the force's line of action [1].

b) $\tau = 0.40 \times 60 \times \sin 30°$ [1] = 12 N m [1].

c) −12 N m [1].

P22. A 25 N force acts 0.32 m from a pivot at 60° to the radius. Calculate its torque magnitude.

Answer

$\tau = rF \sin\theta = 0.32 \times 25 \times \sin 60° = 6.93$ N m ≈ 6.9 N m.

P23. A wheel's angular velocity increases from 2.0 rad/s to 14 rad/s in 3.0 s. Its moment of inertia is 0.75 kg m².

a) Calculate the angular acceleration. [2]

b) Calculate the net torque. [2]

Answer

a) $\alpha = (14 - 2.0)/3.0 = 4.0$ rad/s² [2].

b) $\tau_{net} = 0.75 \times 4.0 = 3.0$ N m [2].

P24. A wheel has a 9.0 N m counterclockwise torque and a 3.0 N m clockwise torque. Its moment of inertia is 2.0 kg m². Use counterclockwise as positive.

a) Calculate the signed net torque. [2]

b) Calculate the angular acceleration. [2]

Answer

a) $\tau_{net} = 9.0 - 3.0 = +6.0$ N m [2].

b) $\alpha = 6.0/2.0 = +3.0$ rad/s² [2].

P25. Two 0.20 kg masses form the ends of a 0.80 m baton. Treat them as point masses and neglect the rod.

a) Calculate I about an axis through the midpoint, perpendicular to the rod. [2]

b) Calculate I about a parallel axis through one end mass. [2]

c) For the same applied net torque, which axis gives greater angular acceleration? State the factor. [1]

Answer

a) $I = 2(0.20)(0.40^2)$ [1] = 0.064 kg m² [1].

b) $I = 0.20(0^2) + 0.20(0.80^2)$ [1] = 0.128 kg m² [1].

c) Midpoint axis gives twice the angular acceleration [1], because its I is half as large.

P26. Two 0.40 kg point masses are each 0.25 m from a fixed central axis. A net torque of 0.50 N m acts on the system. Neglect the connecting rod.

a) Calculate the total moment of inertia. [2]

b) Calculate the angular acceleration. [2]

Answer

a) $I = 2 \times 0.40 \times 0.25^2 = 0.050$ kg m² [2].

b) $\alpha = 0.50/0.050 = 10$ rad/s² [2].

P27. A fixed net torque acts on a system of point masses. All distances from the fixed axis are doubled while the masses stay unchanged.

a) Determine the factor by which the moment of inertia changes. [2]

b) Determine the factor by which the angular acceleration changes. [2]

c) Evaluate the claim that moving the masses outward makes the system accelerate faster. [2]

Answer

a) $I_{new} = \Sigma m(2r)^2 = 4I_{old}$ [2].

b) $\alpha_{new} = \tau_{net}/(4I_{old}) = \alpha_{old}/4$ [2].

c) The claim is incorrect: increasing radius raises I and reduces α for fixed torque [2].

P28. A door is hinged along one edge. Two students apply equal forces perpendicular to the door, one near the hinges and the other near the handle.

a) Explain which force produces the larger torque. [2]

b) Suggest two ways to maximise the torque for the same force magnitude. [2]

Answer

a) The force near the handle gives a larger torque because its lever arm is larger: $\tau = rF$ [2].

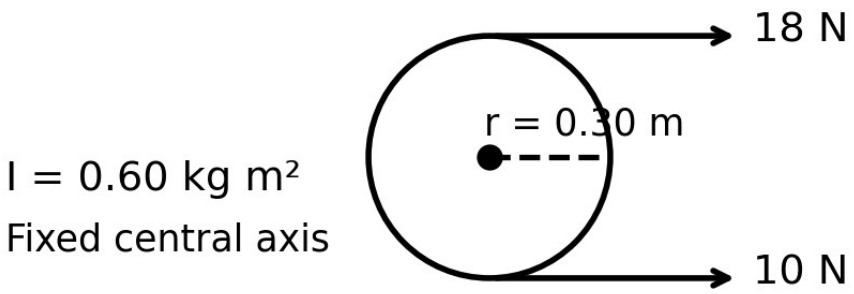
b) Apply the force farther from the hinges [1] and perpendicular to the radius/door [1].

P29. Two tangential forces act on the disk as shown.

a) Calculate the signed net torque and angular acceleration. [2]

b) A student doubles both force radii and keeps both forces unchanged, using a similar disk of the same mass but twice the radius. Evaluate the claim that angular acceleration doubles. Use $I \propto r^2$. [2]

c) Design a change to the distribution of mass that increases angular acceleration at the original net torque. Explain using an equation. [2]



Answer

a) $\tau_{net} = 0.30(10 - 18) = -2.4$ N m [1]; $\alpha = -2.4/0.60 = -4.0$ rad/s² [1].

b) Torque doubles [1]; I quadruples, so α halves. The claim is rejected [1].

c) Move mass closer to the central axis [1]; this reduces $I = \Sigma mr^2$, so $\alpha = \tau_{net}/I$ increases in magnitude [1].

P30. A motor applies a 5.0 N m driving torque to a rotor with I = 0.50 kg m². A constant 1.0 N m frictional torque opposes the motion. The rotor starts from rest.

a) Calculate the net torque. [1]

b) Calculate the angular acceleration. [2]

c) Calculate its angular velocity after 4.0 s. [2]

d) Suggest a change that increases angular acceleration without changing the rotor's moment of inertia or driving torque. [1]

Answer

a) $\tau_{net} = 5.0 - 1.0 = 4.0$ N m [1].

b) $\alpha = 4.0/0.50 = 8.0$ rad/s² [2].

c) $\omega_f = 0 + 8.0 \times 4.0 = 32$ rad/s [2].

d) Reduce opposing frictional torque, raising net torque and hence $\alpha = \tau_{net}/I$ [1].